First degree.

Consider an equation $y' + a(t) \cdot y = 0$. Let's solve this.

If $y(t_0) = 0$ for some $t_0 \in I$, then since $\tilde{y} \equiv 0$ is a solution s.t $\tilde{y}(t_0) = 0$, the uniqueness theorem gives $y \equiv 0$. Now, assume $y > 0$ for all $t \in I$.

Then, we obtain $\frac{y'}{y} = (\log y)' = -a(x)$, so

$$\log y = -\int_{t_0}^{t} a(x)\,dx \Rightarrow y = \exp\left(-\int_{t_0}^{t} a(x)\,dx\right)$$

Non-Homogeneous Case

Consider an equation $y' + a(t) \cdot y = b(t)$.

We already know the Homogeneous Solution

$$y_H = \exp\left(-\int_{t_0}^{t} a(x)\,dx\right).$$

And, the particular solution is

$$y_p(t) = y_H(t) \cdot \int_{t_0}^{t} \frac{b(x)}{y_H(x)}\,dx$$

Therefore, the general solution is

$$y(t) = y_H(t) \cdot \left[C + \int_{t_0}^{t} \frac{b(x)}{y_H(x)}\,dx\right]$$

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Consider the equation

$$y'' + \frac{1}{x} \cdot y' - \frac{1}{x^2} \cdot y = 0 \quad \text{for } x > 0.$$

Find the general solution of the ODE.

Proof. Observe that: let $y = x^r$ for some r. Then,

$$r \cdot (r-1) \cdot x^{r-2} + \frac{1}{x} \cdot r \cdot x^{r-1} - \frac{1}{x^2} \cdot x^r$$

$$= x^{r-2}\left(r(r-1) + r - 1\right) = x^{r-2} \cdot \left(r^2 - 1\right)$$

Therefore, since $r^2 - 1 = (r-1)(r+1)$,

$y = x$ and $y = \frac{1}{x}$ are solutions,